

Euler-Bernoulli Beam theory:- / Simple Beam theory, Derivation of the Euler-Bernoulli Beam and Bending stress formulae:- (1)

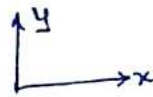
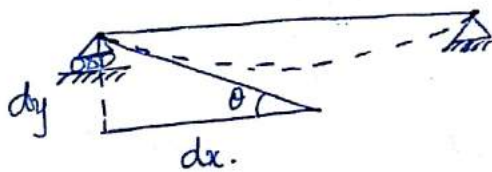
Assumptions made:-

- (a) The beam must be long, relative to its depth and width.
- The beam is a structure which has one of its dimensions larger than the other two. (Plane sections remain plane).
- (b) The Cross-section of the beam remains constant along its length.
- (c) The beam is symmetrical about the Y-Y axis, hence the result is there is no twisting or torsion occurring.
- (d) Deflections / deformations are small. (Poisson ratio effects are ignored).
- (e) The material the beam is made up of is isotropic and obey's Hooke's law. (linearly elastic beam).

The beam theory involves the following considerations:-

- (a) Type of material
- (b) The way the beam deforms
- (c) The geometry of the beam
- (d) The internal equilibrium.

Curvature:- Denoted by the letter 'K', is the rate of change of the beam slope.

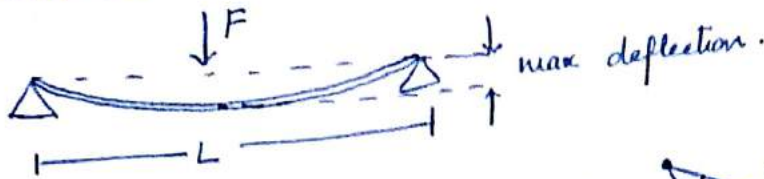


Note: θ is given in radians.

Consider the deflected beam shown above.
The slope given by $dy/dx = \tan \theta$, but when the deflections are small, we can approximate, $dy/dx \approx \theta$.
Hence the slope becomes θ and K which is the rate of change of slope (θ).

The Euler Bernoulli, Beam theory defines the relationship between the beam's deflection and the applied load. The curve defines the beam's deflection.

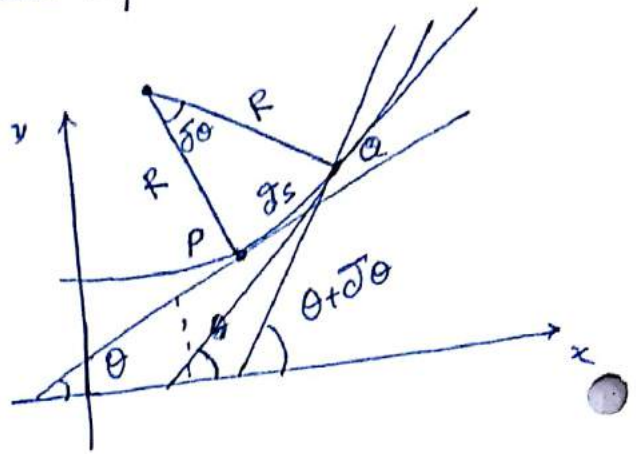
Let us now consider the beam shown below:-



$\delta s \rightarrow$ distance between P and Q

$\delta \theta \rightarrow$ change in angle.

$\theta \rightarrow$ slope.



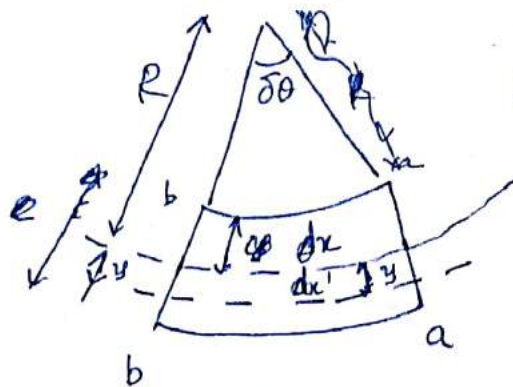
hence, we can write the change in slope, to be.

$$k = \frac{\theta + \delta \theta - \theta}{\delta s} = \frac{d\theta}{ds}$$

Assuming plane sections remain plane, the Radius R is that of a circle. Hence,

$$R \cdot d\theta = ds \quad \therefore k = \frac{d\theta}{ds} = \frac{1}{R} \quad \left(\text{This is the relationship b/w Radius } R, \text{ slope and curvature} \right)$$

Assume, a section of the deflected beam shown below.



At a distance y - from the neutral axis, we need to the strain ϵ .

$$\epsilon = \frac{\text{change in } L}{L} = \frac{dx' - dx}{dx} \quad \text{--- (1)}$$

$$\text{Again, } R \cdot d\theta = dx \quad \text{--- (2)}$$

$$(R+y) d\theta = dx' \quad \text{--- (3)}$$

(2) & (3) in (1).

$$\frac{((R+y) - R) d\theta}{R d\theta} = \epsilon$$

$$\frac{R d\theta + y d\theta - R d\theta}{R d\theta} = \epsilon$$

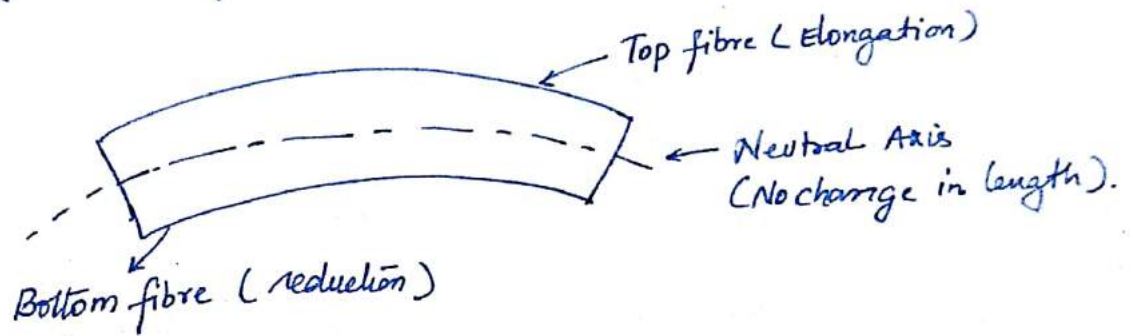
$$\frac{y}{R} = \epsilon \quad \boxed{\epsilon = k y}$$

NEUTRAL AXIS:

(2)

When a beam bends, we have the top fibre going under elongation and bottom fibre going under compression.

When we come closer to the geometrical Centroid of the beam we find a longitudinal fibre which does not undergo a change in length. This fibre will then be considered as the neutral axis.

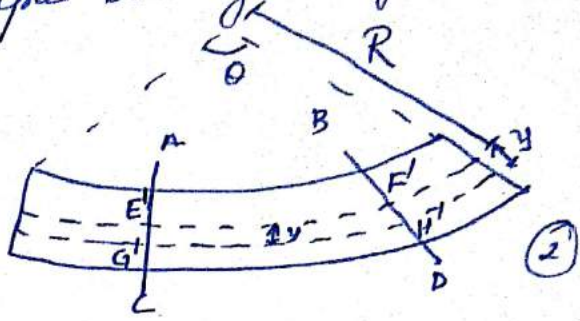
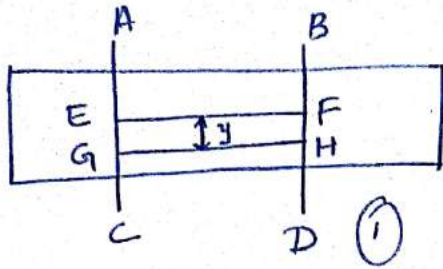


Hence, we make the following assumptions:-

- ① Neutral Axis does not experience any change in length.
- ② All cross sections of the beam remain plane and perpendicular to the neutral axis.
- ③ Because of ① the neutral axis will not experience any axial force.

Simple Bending Theory:-

A theory which deals with the finding stresses at a section due to pure moment is called simple bending theory. (3)



Since EF is the neutral axis, we can say $EF = E'F'$

$$E'F' = EF = R\theta. \quad \text{--- (1)}$$

Let us find the strain in fibre $G'H'$ which is

$$\textcircled{2} - \epsilon_{GH} = \frac{G'H' - GH}{GH}, \text{ from fig (1) we can say,}$$

Substituting (1) in (2) we get, as $EF = GH$

$$\epsilon_{GH} = \frac{G'H' - R\theta}{R\theta}, \text{ But from the fig (2) we can get } G'H' = (R+y)\theta$$

$$\text{hence, } \epsilon_{GH} = \frac{(R+y)\theta - R\theta}{R\theta} = \frac{R\theta + y\theta - R\theta}{R\theta} = \boxed{\frac{y}{R} = \epsilon_{GH}} \quad \text{--- (3)}$$

$\therefore$ Strain in the fibre $G'H' = \frac{y}{R}$, where $\left\{ \begin{array}{l} y \rightarrow \text{distance of the} \\ \text{fibre from neutral} \\ \text{axis} \end{array} \right.$

and R is the radius of curvature of the beam.

from hook's law we can derive $\sigma_b = E\epsilon$.

Hence, $E = \frac{\sigma_b}{\epsilon}$ --- (4), Substituting (4) in (3), we get.

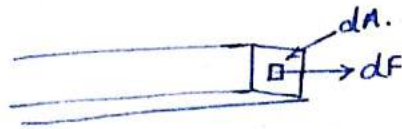
$$\boxed{\frac{y}{R} = \frac{\sigma_b}{E}}$$

when σ_b is the bending stress and E is the modulus of Elasticity.

$\hookrightarrow \textcircled{5}$

Euler-Bernoulli Equation and the bending equation: (4)

Let us consider the beam and a small element with area dA as shown in fig.



We know the stress developed in that area dA would be σ due to the applied force dF . Hence,

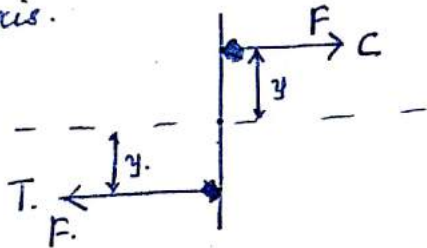
$$\sigma = dF/dA. \text{ or } dF = \sigma dA.$$

Integrating we get, $F = \int_A \sigma dA$. — (1)

from the previous equation we know $\sigma = \frac{E}{R} \cdot y$. Replacing in eqn. (1)

$$\text{So } F = \frac{E}{R} \int_A y \cdot dA. \text{ — (2)}$$

When the beam bends, the bending induces a couple/moment about the neutral axis.



Taking the moments.

$$F \cdot y + F \cdot y = M. \text{ — (3)}$$

Replacing F in equation (3) by (2) we get,

$$M = \frac{E}{R} \int_A y^2 \cdot dA = M. \text{ — (4)}$$

We know $\int_A y^2 \cdot dA = I$ which is Second Area moment or otherwise called the ^{Area} moment of inertia.

Hence equation (4) becomes

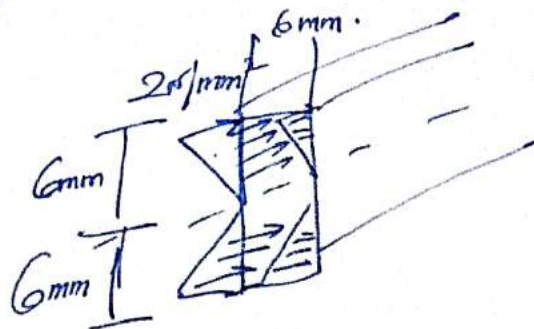
$$\boxed{\frac{M}{I} = \frac{E}{R}} \text{ which is the Euler-Bernoulli equation for bending.}$$

Hence the bending equation becomes.

$$\boxed{\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}}$$

Beams Subjected to axial loads:-

A beam has a rectangular cross section and is subjected to stress distribution as shown.



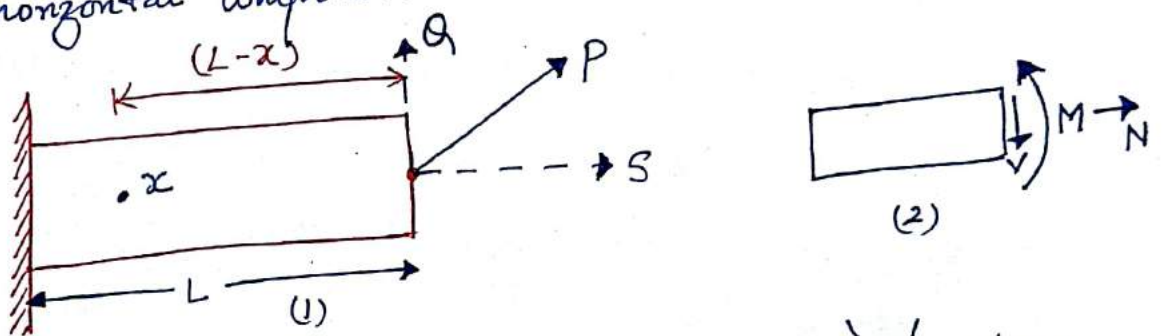
Determine internal moment M at the section caused by the stress distribution.

BEAMS WITH Axial loads & Combined loads

(5)

Structural Members are often subjected to Simultaneous axial and bending loads. Especially in aircraft frames if the members are slender, i.e. the member is very long when compared to its cross-sectional dimensions. The combined stresses (Bending and Axial) can be obtained by principle of superposition.

Consider the cantilever beam shown. A inclined force P acts on the end of the beam. This load can be resolved as vertical & horizontal component.



These loads now produce, a bending moment M , Shear force V and axial force N . through the beam as shown in (2). Now, at a distance x from the support,

$$\text{Moment } M = Q \times (L-x)$$

$$V = -Q$$

and the normal force is $N = S$.

Now, Since both the axial and Bending moment M produce stresses we need to combine them.

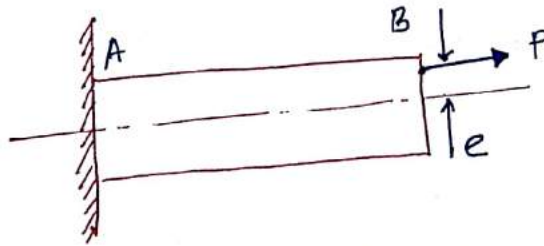
$$\sigma_N = N/A \quad \text{and} \quad \sigma_B = -My/I.$$

Thus, the combined stress would be, $\sigma = \frac{N}{A} - \frac{M \cdot y}{I}$.

Beams Subjected to eccentric Axial loads :-

(6)

An eccentric load is an axial force that does not act through the Centroid of the Cross-section.

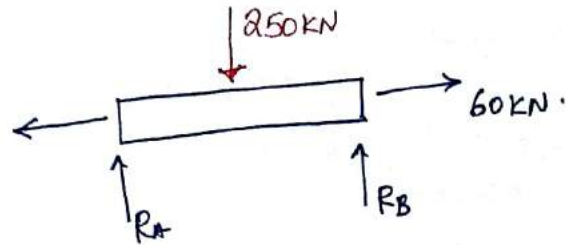


A tensile load P acts on the beam at a distance ' e ' from the Centroid. This distance ' e ' is called eccentricity. This eccentric load ' P ' is statically equivalent to a normal force ' P ' acting through the Centroid and a moment about the Centroid, which is ' $P \cdot e$ ' acting clockwise. Hence the normal stress at any point in the Cross section is.

$$\sigma = \frac{P}{A} + \frac{P \cdot e \cdot y}{I}$$

Combined stresses Examples:-

2m long 51mm dia, standard pipe is loaded axially in tension with a load of 60 kN. The bar also is a simply supported beam, transversely loaded with a load of 250 kN. Calculate the maximum stress in the member.



C/S Area =

Combined stress = Axial stress + Bending stress.

$$\sigma_C = \sigma_A + \sigma_B$$

$$\sigma_{\text{Axial}} = P/A = \frac{60 \times 10^3}{\frac{\pi d^2}{4}} = 29.39 \text{ N/mm}^2$$

from Bending Moment Equations:-

$$M = \frac{P \cdot L}{4} \Rightarrow \text{Hence the stress becomes}$$

$$\sigma_B = \frac{M}{Z} \quad \text{where } Z \text{ is Section Modulus.}$$

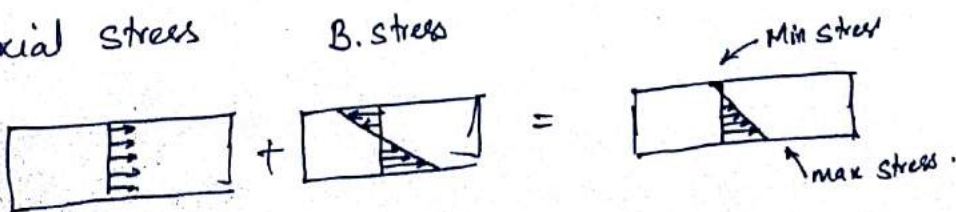
$$Z = \frac{\pi d^3}{32} = 13016.37 \text{ mm}^3$$

$$\sigma_B = \frac{250 \times 10^3 \times 2000}{4 \times 13016.37} = 9603.29 \text{ N/mm}^2$$

$$\text{Combined stress} = 9603.29 + 29.39 = 9632.68 \text{ N/mm}^2$$

Axial stress

B. stress



$$\text{Hence min stress} = 29.39 - 9603.29 = -9574.39 \text{ N/mm}^2$$

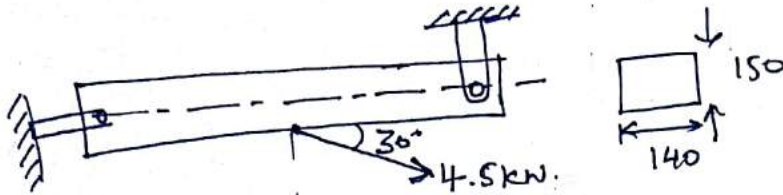
$$29.39 - 9603.29 = -9574.39 \text{ N/mm}^2 //$$

(8)

A Tubular Beam ACB of length $L = 1.5\text{m}$ loaded by an inclined force P at mid length.

$$P = 4.5\text{ kN} \quad d = 140\text{ mm} \quad b = 150\text{ mm}$$

$$A = 12,500\text{ mm}^2 \quad L = 1.5\text{ m} \quad I = 33.86 \times 10^6\text{ mm}^4.$$



$$F_H = 4.5 \cos 30^\circ = 3897\text{ N}$$

$$F_V = 4.5 \sin 30^\circ = 2250\text{ N.}$$

$$\cancel{F_H} = \cancel{F_V} = \cancel{3897\text{ N}}$$

$$\frac{N}{A} = 0.31\text{ N/mm}^2$$

$$\frac{M}{I} = 843.750 \times 10^3 \text{ N/mm}^2$$

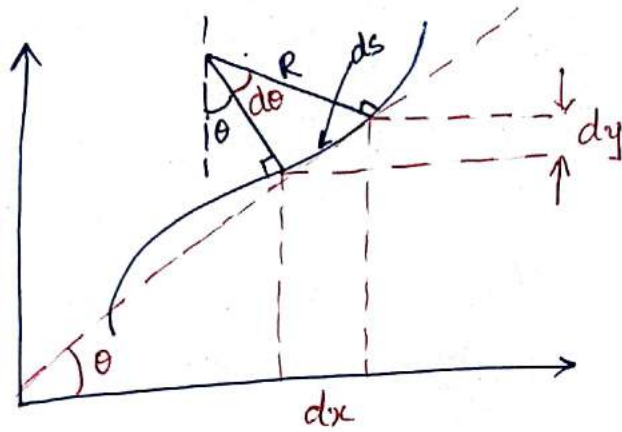
$$My/I = 1.8689\text{ N/mm}^2$$

$$\sigma_{\max}(\text{tension}) = \frac{N}{A} + \frac{My}{I} =$$

$$= 0.31 + 1.8689 = 2.178\text{ MPa.}$$

$$\sigma_{\max}(\text{Compression}) = \frac{N}{A} - \frac{My}{I} = -1.558\text{ MPa}$$

Equation of Elastic Curve:-



From the fig.

$$\tan \theta = \frac{dy}{dx}$$

for small ' θ '.

$$\theta = \frac{dy}{dx}$$

$$ds = R d\theta$$

But for small curvatures $ds = dx$

$$dx = R d\theta$$

$$R = \frac{dx}{d\theta} \quad \text{or} \quad \frac{d\theta}{dx} = \frac{1}{R} \quad \text{Substituting value of } \theta \text{ in this equation.}$$

$$\frac{d}{dx} \left[\frac{dy}{dx} \right] = \frac{d^2y}{dx^2} = \frac{1}{R} \quad \text{--- (1)}$$

Now, from the bending Equation:-

$$\frac{M}{I} = \frac{E}{R} = \frac{\sigma}{y}, \quad \text{taking } \frac{M}{I} = \frac{E}{R} \text{ and substituting in (1)}$$

We get. $\frac{d^2y}{dx^2} = \frac{M}{EI}$, where EI is called the flexural Rigidity.

$$\boxed{EI \frac{d^2y}{dx^2} = M}$$

Equation of the elastic Curve in Bending.

For Additional information:-

$$\text{Slope} = \frac{dy}{dx}, \quad \text{B.M} = EI \cdot \frac{d^2y}{dx^2}, \quad \text{Shear force} = EI \frac{d^3y}{dx^3}$$

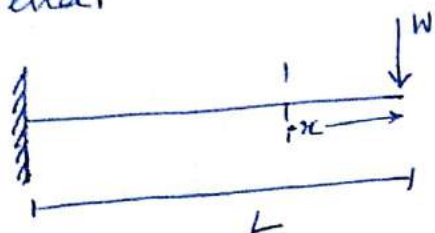
$$\text{and load distribution} = EI \frac{d^4y}{dx^4} //$$

Double integration Method:-

(10)

$$M = EI \frac{d^2y}{dx^2}$$

Case 1 Cantilever Subjected to Concentrated load at free end.



$$M(x) = -Wx$$

$$EI \frac{d^2y}{dx^2} = -Wx$$

$$EI \frac{dy}{dx} = -\frac{Wx^2}{2} + C_1 \quad \text{--- (1)}$$

Ex. $y \neq$ Slope at ~~free~~ ^{fixed} end is zero. at $x=L$ $dy/dx = 0$

hence $C_1 = +\frac{WL^2}{2}$ [Replacing in (1)]

$$EI \frac{dy}{dx} = -\frac{Wx^2}{2} + \frac{WL^2}{2}$$

Integrating. $EI y = -\frac{Wx^3}{6} + \frac{WL^2x}{2} + C_2 \quad \text{--- (2)}$

at $x=L$, $y=0$.

$$C_2 = -\frac{WL^2 \cdot L}{2} + \frac{WL^3}{6} = -\frac{WL^3}{2} + \frac{WL^3}{6}$$

$$C_2 = -WL^3 \left[\frac{1}{2} - \frac{1}{6} \right] = -\frac{WL^3}{3} \quad \text{--- (3)}$$

Replacing in Eqn. (2)

$$EI y = -\frac{Wx^3}{6} + \frac{WL^2x}{2} - \frac{WL^3}{3}$$

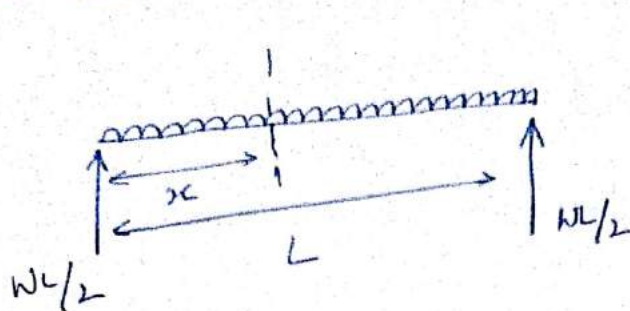
at free end $x=0$.

$$EI y = -\frac{WL^3}{3}$$

$$y = -\frac{WL^3}{3EI} \quad (\text{Negative symbols indicate downward bending}).$$

Simply Supported Beam Subjected to Uniformly distributed load.

(11)



$$M_x = \frac{WLx}{2} - \frac{Wx^2}{2}$$

$$EI \frac{d^2y}{dx^2} = \frac{WL}{2}x - \frac{Wx^2}{2}$$

Integrating:- $EI \frac{dy}{dx} = \frac{WL}{2} \frac{x^2}{2} - \frac{Wx^3}{6} + C_1$

$dy/dx = 0$ at $x = L/2$.

$$0 = \frac{WL}{2} \cdot \frac{L^2}{4} - \frac{W}{6} \frac{L^3}{8} + C_1 \Rightarrow C_1 = \frac{WL^3}{48} - \frac{WL^3}{16}$$

$$EI \frac{dy}{dx} = \frac{WL}{2} \frac{x^2}{2} - \frac{Wx^3}{6} + WL^3 \left[\frac{1}{48} - \frac{1}{16} \right] = \frac{WL}{2} \frac{x^2}{2} - \frac{Wx^3}{6} + \frac{WL^3}{24}$$

Integrating on both sides.

$$EI \cdot y = \frac{WL \cdot x^3}{12} - \frac{Wx^4}{24} - \frac{WL^3 x}{24} + C_2$$

at $x=0$ $y=0$, $C_2=0$. at $x=L/2$ deflection is maximum.

$$EI \cdot y = \frac{WL}{12} \left(\frac{L}{2} \right)^3 - \frac{W}{24} \left[\frac{L}{2} \right]^4 - \frac{WL^3}{24} \left[\frac{L}{2} \right]$$

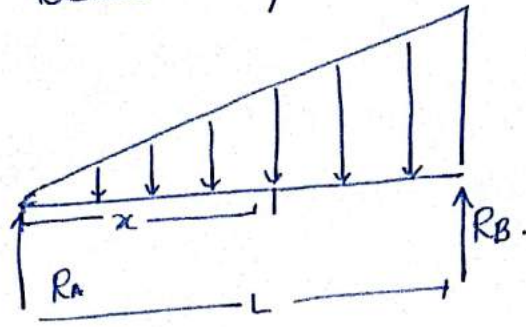
$$EI \cdot y = \frac{WL^4}{96} - \frac{WL^4}{384} - \frac{WL^4}{48} = \frac{WL^4}{96} \left[\frac{1}{1} - \frac{1}{4} - \frac{2}{1} \right]$$

Simplifying $EI \cdot y = \frac{WL^4 \cdot 5}{384}$

$$\text{max } y = \frac{WL^4 \cdot 5}{384 EI}$$

Simply Supported Beam Subjected to Uniformly Varying Load.

(12)



$$R_A = W \cdot \frac{L}{3} \cdot \frac{1}{2} = \frac{WL}{6}$$

$$R_A \cdot x = \frac{WLx}{6}$$

Look at SFD, BMD for better clarity.

$$M_x = \frac{WLx}{6} - \frac{1}{2}x \cdot \frac{Wx}{L} \cdot \frac{x}{3} = \frac{WLx}{6} - \frac{Wx^3}{6L}$$

$$\therefore EI \cdot \frac{d^2y}{dx^2} = \frac{WLx}{6} - \frac{Wx^3}{6L}$$

$$EI \cdot \frac{dy}{dx} = \frac{WLx^2}{12} - \frac{Wx^4}{24L} + C_1$$

$$EI \cdot y = \frac{WLx^3}{36} - \frac{Wx^5}{120L} + C_1x + C_2$$

Applying Boundary Conditions:-

at $x=0$, $y=0$, $C_2=0$

at $x=L$, $y=0$

$$0 = \frac{WL^3}{36} - \frac{WL^5}{120L} + C_1L$$

$$C_1L = -\frac{WL^3}{36} + \frac{WL^4}{120} \Rightarrow C_1 = WL^3 \left[\frac{1}{120} - \frac{1}{36} \right]$$

Substituting back and solving for max deflection we get
Using $dy/dx=0$ at max deflection, we get.

$$y_{max} = -0.006523 \frac{WL^4}{EI}$$

$$\text{or } y_{max} = \frac{WL^4}{154EI} \text{ (Approx) at } \underline{\underline{0.5193L}}$$

Statically indeterminate Beam's

(13)

Statically indeterminate Beam's are beam's whose reactions exceed the number of independent equations of equilibrium.

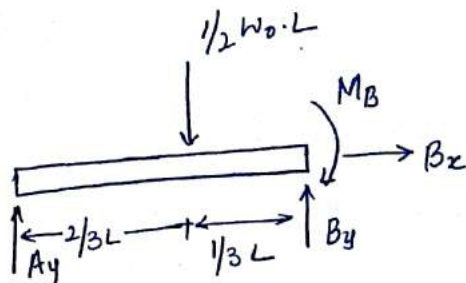
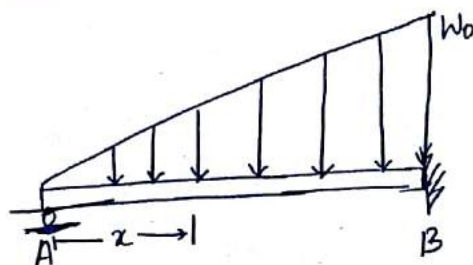
Since, the reactions cannot be determined by statics alone, the beams are said to be statically indeterminate.

In this case we will adopt a method where we will supplement the equilibrium equations with compatibility equations and force-displacement equations.

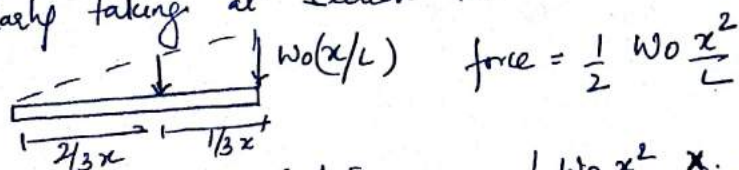
These excess reactions are called redundants. (static redundants)

Statically Indeterminate Beams - Method of Integration.

The beam shown below is subjected to a distributed loading. Determine reactions at A, assuming EI is constant.



Similarly taking at section x .



Hence moment at section $x = \frac{1}{2} w_0 \frac{x^2}{L} \cdot x \cdot \frac{1}{3} \cdot x = -\frac{1}{6} w_0 \frac{x^3}{L} + A_y \cdot x$

Substituting it to, $EI \cdot \frac{d^2 y}{dx^2} = M = -\frac{1}{6} w_0 \frac{x^3}{L} + A_y \cdot x$

find slope by integrating.

$$EI \cdot \frac{dy}{dx} = -\frac{1}{6} \frac{w_0 x^4}{4L} + \frac{A_y \cdot x^2}{2} = \frac{A_y x^2}{2} - \frac{1}{24} w_0 x^4 + C_1$$

find deflection by integrating again.

$$EI \cdot y = +\frac{A_y x^3}{6} - \frac{1}{120} w_0 x^5 + C_1 x + C_2$$

from the equation:-

(1.4)

$$EI \cdot y = \frac{1}{6} A_y x^3 - \frac{1}{120} w_0 \frac{x^5}{L} + C_1 x + C_2$$

There are three Unknowns, A_y , C_1 and C_2 . which can be determined from the boundary conditions.

$$x=0, y=0, x=L \text{ and } dy/dx=0, x=L, y=0.$$

Taking the first set of boundary conditions.

$$x=0, y=0.$$

$$\text{We get, } 0 = 0 - 0 + 0 + C_2 \quad | \quad C_2 = 0.$$

Now, with the second set of boundary conditions.

$$x=L \quad dy/dx=0.$$

Now taking the equation of slope

$$EI \cdot dy/dx = \frac{1}{2} A_y x^2 - \frac{1}{24} w_0 \frac{x^4}{L} + C_1$$

Applying boundary conditions.

$$0 = \frac{1}{2} A_y L^2 - \frac{1}{24} w_0 \frac{L^4}{L} + C_1$$

$$0 = \frac{A_y L^2}{2} - \frac{w_0 L^3}{24} + C_1 \rightarrow C_1 = \frac{w_0 L^3}{24} - \frac{A_y L^2}{2} \quad \text{--- (2)}$$

Now let us solve for A_y by applying the third set of Boundary conditions.

$$x=L, y=0.$$

$$0 = \frac{A_y L^3}{6} - \frac{1}{120} w_0 L^4 + \frac{A_y L^3}{2} + \frac{w_0 L^4}{24} + 0$$

$$0 = A_y L^3 \left[\frac{1}{6} - \frac{1}{2} \right] - w_0 L^4 \left[\frac{1}{24} - \frac{1}{120} \right]$$

$$A_y L^3 \left[\frac{4}{24} \right] = w_0 L^4 \left[\frac{1}{30} \right] \rightarrow A_y L^3 \left[\frac{1}{3} \right] = w_0 L^4 \left[\frac{1}{30} \right]$$

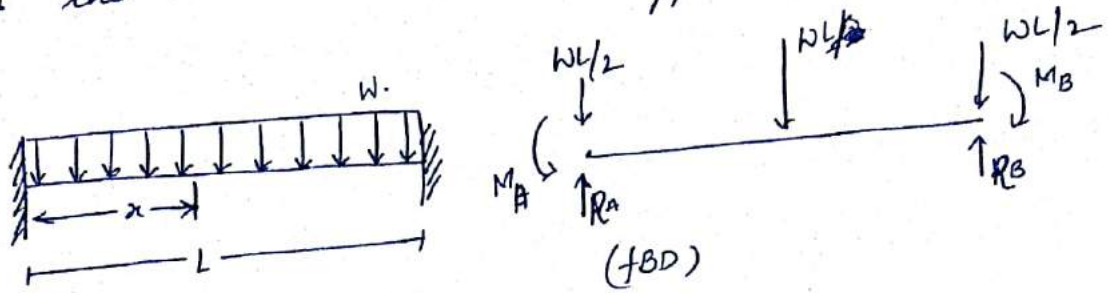
$$A_y L^3 = w_0 L^4 \left[\frac{1}{10} \right] = w_0 L \left[\frac{1}{10} \right]$$

$$\boxed{A_y = w_0 L \left[\frac{1}{10} \right]}$$

$$\boxed{C_1 = \frac{1}{120} w_0 L^3}$$

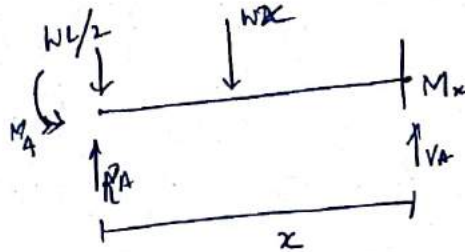
$$\boxed{C_2 = 0}$$

The beam shown is a fixed beam, fixed at both ends, (15) and is subjected to Uniform loading as shown, Determine the reactions at the supports. (Neglect axial loads).



$$M_B = M_A =$$

$$M_A = \frac{WLx}{2}$$



From Statics. $\frac{WL}{2} = R_A = R_B$

Taking moments about x-section plane.

$$M_x = R_A x - wx \cdot \frac{x}{2} = \frac{WL}{2} \cdot x - \frac{wx^2}{2} - M_A$$

Applying equation of elastic curve.

$$EI \frac{d^2y}{dx^2} = \frac{WLx}{2} - \frac{wx^2}{2} - M_A$$

Integrating. $EI \frac{dy}{dx} = \frac{WLx^2}{4} - \frac{wx^3}{6} - M_A x + C_1$

Integrating again,

$$EI \cdot y = \frac{WLx^3}{12} - \frac{wx^4}{24} - \frac{M_A x^2}{2} + C_1 x + C_2$$

Applying Boundary Conditions :-
 $x=0, y=0$ & $x=L, y=0$ & $x=0, dy/dx=0$.

$$E.I. y = \frac{WLx^3}{12} - \frac{Wx^4}{24} - \frac{M_A \cdot x^2}{2} + C_1 x + C_2$$

(16)

Applying $x=0, y=0$.

$$0 = 0 - 0 - 0 + 0 + C_2 \quad \text{Hence } \boxed{C_2 = 0}$$

Applying, $x=L, y=0$

$$0 = \frac{WL^4}{12} - \frac{WL^4}{24} - \frac{M_A \cdot L^2}{2} + C_1 L \quad \text{--- (2)}$$

Applying $x=0, dy/dx=0$,

$$EI dy/dx = \frac{WLx^2}{4} - \frac{Wx^3}{6} - M_A x + C_1 \quad \text{--- (3)}$$

$$0 = 0 - 0 - 0 + C_1, \quad \boxed{C_1 = 0}$$

From equation (2) we get,

$$+\frac{M_A L^2}{2} = \frac{WL^4}{12} - \frac{WL^4}{24}$$

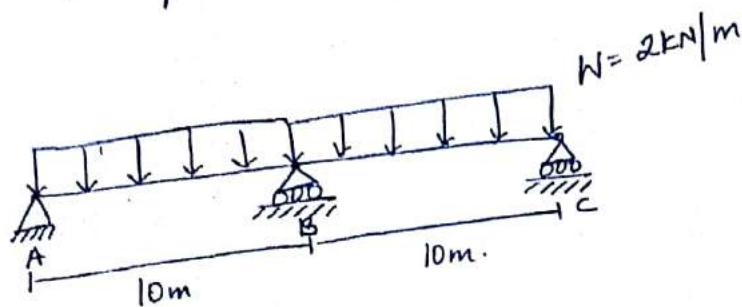
$$\frac{M_A L^2}{2} = \frac{WL^4}{24}$$

$$\rightarrow \boxed{M_A = \frac{WL^2}{12}}$$

Hence, we can solve, the Unknown's w.r.t. the loading and the length of beam.

Method of Superposition for a statically indeterminate.

(17)

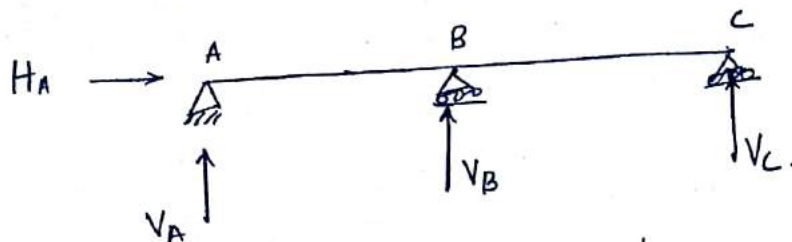


$EI = \text{Constant.}$

So, now let us calculate the degree of redundancy.

Number of equilibrium equations = $\begin{cases} \sum F_x = 0 \\ \sum F_y = 0 \\ \sum M = 0 \end{cases}$ Hence we have 3 equilibrium equations.

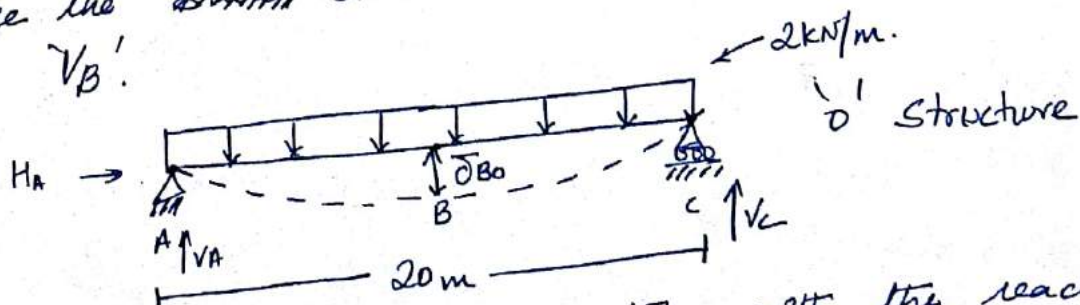
The number of reactions for the structure is,



The number of unknowns are 4 hence we have $1(4-3)$ redundant reactions or the order of redundancy is 1.

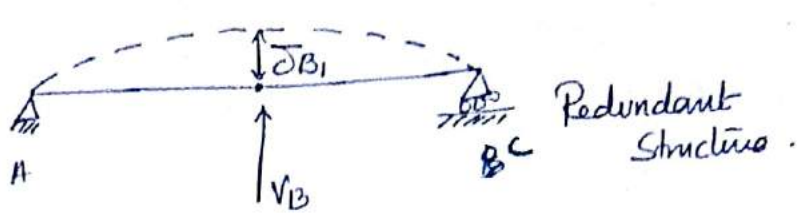
Note:- Method of Superposition should be used when the order of redundancy is 1 or maximum 2. i.e. the structure must be stable if the redundancy is removed.

Now, to make the ~~stable~~ structure determinant, let us remove the reaction V_B .



Now let us introduce another structure with the reaction V_B . And if we superimpose the two structures, we should be able to get all the reactions in our initial structure. We can write compatibility equations for the structures.

'1' structure



(18)

Compatibility equations:-

$$\delta_{B_0} \text{ at } (10\text{m}) + \delta_{B_1} \text{ at } (10\text{m}) = 0 \quad \text{--- (1)}$$

i.e. the deformation of the '0' structure at 'B' and an addition of the '1' structure at 'B' should give me the overall deflection at 'B'.

Now, let us calculate the deformation at B for structure '0' and structure '1'.

for a simply supported beam, the deflection is given by $\frac{+5WL^4}{384EI}$, so taking the bending as negative and taking hogging as positive, we can write equation (1) as,

$$\frac{-5WL^4}{384EI} + \frac{V_B L^3}{48EI} = 0 \quad \text{and equating it to zero.}$$

Now for structure 1 we know the deflection for a concentrated point loading acting in the middle is given by.

$$\frac{V_B L^3}{48EI}$$

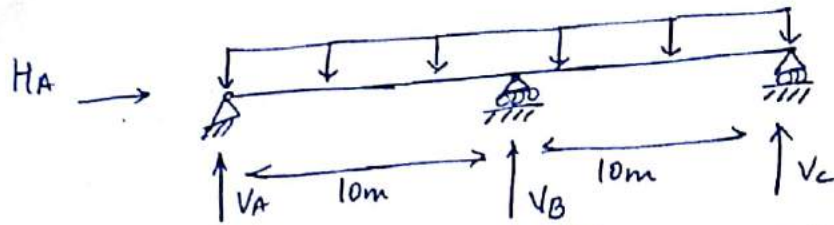
$$\frac{-5WL^4}{384EI} + \frac{V_B L^3}{48EI} = 0 \quad \rightarrow \quad \frac{-5WL}{384} + \frac{V_B}{48} = 0$$

Now, substituting the values of W and L

$$\frac{-5 \times 20 \times 2}{384} + \frac{V_B}{48} = 0, \text{ solving we get, } \underline{V_B = 25 \text{ kN.}}$$

Now, we can know by sign conventions that V_B acts upward.

Now, let us, solve for the other reactions from the initial structure. (19)



Now, taking moments about (A), we get,

$$-V_C \times 20 - V_B \times 10 + W.L(10) = 0 \rightarrow [\sum M_A = 0]$$

$$\Rightarrow -V_C(20) - V_B(10) + 20 \cdot 20 \cdot 10 = 0$$

Solving by substituting $V_B = 25 \text{ kN}$ we get.

$$V_C = 7.5 \text{ kN.}$$

Now, taking $\sum F_y = 0$.

$$-40 + V_A + V_B + V_C = 0$$

$$V_A = 7.5 \text{ kN.}$$

Note: All reactions are acting upwards.

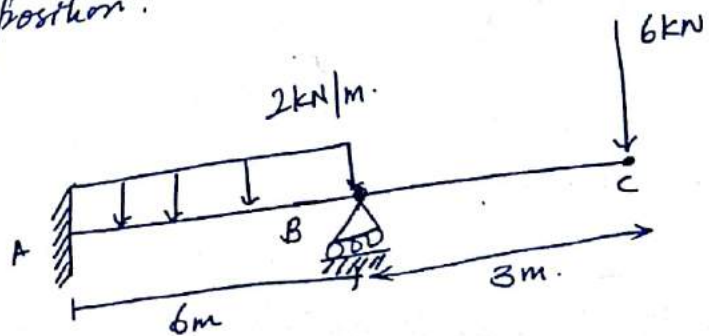
Now, taking $\sum F_x = 0$

$$H_A = 0.$$

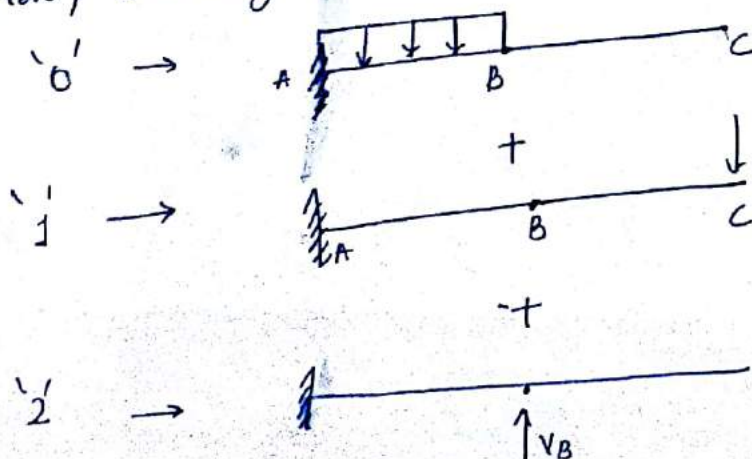
Hence, we are able to solve the statically indeterminate beam by the principle of superposition.

Assignment :-

Solve for all reactions :-



Now, Breaking into 3 structures. ∴



Sum of these three structures will give me the final structure.

Deflection at (B) should be zero, which is at 6m.

$$\delta_B = 0 = -\delta_0 - \delta_1 + \delta_2 \quad [\delta_2 \text{ is positive because of upward deflection}]$$

L (1)

'0' structure is a UDL on cantilever. [neglecting the additional section till C].

$$\text{So, } -\delta_0 = -\frac{WL^4}{8EI} \quad \text{or} \quad \frac{WL^2(L-x)}{6EI}$$

'1' structure is a cantilever beam and deflection at section 'X' is given by $\frac{WL^2(3L-x)}{6EI}$ and

'2' structure is a cantilever beam subjected to end load. deflection is given by $WL^3/3EI$.

Hence substituting in (1) we get!

$$\text{deflection at 'B' } \delta_B = -\frac{WL^4}{8EI} - \frac{WL^2(3L-x)}{6EI} + \frac{WL^3}{3EI}$$

Note: The values of 'L' keep changing for the three structures.

$$\delta_B = \frac{-2(6)^4}{8EI} - \frac{6(6)^2[3(9)-6]}{6EI} + \frac{V_B \cdot 6^3}{3EI} = 0$$

Solving, we get.

$$\boxed{V_B = 15 \text{ kN}} \quad \text{positive indicates, 15 kN acting upward.}$$

Solving for the other reactions. from the actual structure, we can take moments about the fixed end.

$$\sum M_A = 0 = (-6 \times 9) + (V_B \times 6) - \left(\frac{2(6)}{2} \times 3 \right) + M_A \quad \text{--- (1)}$$

~~Solving for V_B~~ Hence, $\sum M_A = 0 = M_A$.

Now, the other equation of equilibrium,

$$\sum F_y = 0.$$

$$V_A + V_B - 12 - 6 = 0 \quad \text{--- (2)}$$

$$V_B = 15 \text{ kN.}$$

$$V_B = 3 \text{ kN.}$$

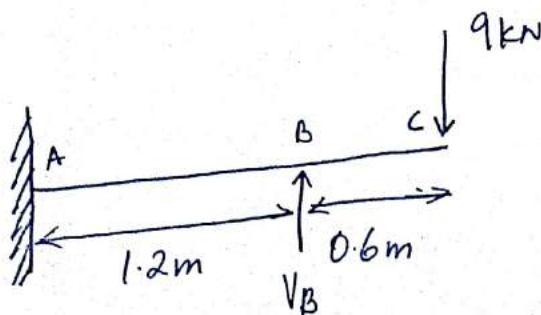
and $\sum F_x = 0$, we get.

$$H_A = 0.$$

Hence the ~~for~~ unknown reactions can be solved by this method of superposition.

~~Summary of the method~~

Additional Questions for practice:-

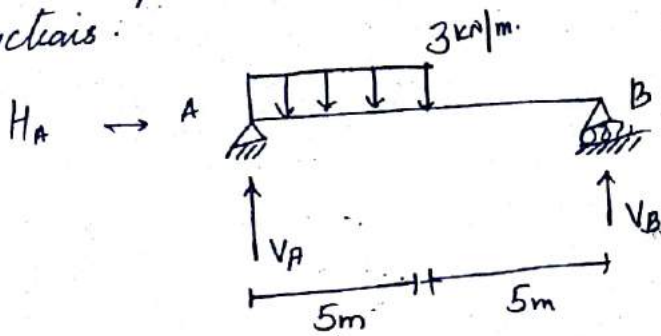


Singularity functions:- (statically indeterminate structure).

(22)

- ① Describe the use of Singularity functions for deflection calculations.
- ② Define singularity functions for various loadings.
- ③ Explain process for using singularity functions for beam deflections.

for the slope and deflection equations using singularity functions.



$$\sum F_x = 0$$

$$H_A = 0$$

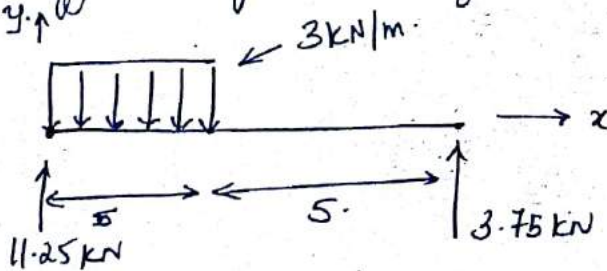
$$\sum M_A = 0, \quad -V_B \times 10 + (3 \times 5) \left(\frac{5}{2} \right) = 0,$$

Hence, we get. $V_B = 3.75 \text{ kN}$.

$$\sum F_y = 0, \quad V_A + V_B - 3(5)$$

$$V_A = 11.25 \text{ kN}.$$

Drawing Only the forces, acting on the beam.



The load distribution is given by

$$\frac{d^4y}{dx^4} = \frac{w}{EI}(x) \text{ load over span } \cdot x.$$